

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I Beffy A Shanika(192511387) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: beffy

Assessment Tasks

Class Test

1. Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.
2. Explain the 5Vs of Big Data with one practical example for any two Vs.
3. Differentiate structured, semi-structured, and unstructured data with examples.
4. Explain the role of distributed file systems in big data processing.
5. List any three major security concerns in big data platforms and briefly explain each.
6. What are data appliances and integration tools? Explain their role in big data workflows.
7. Discuss two major challenges organizations face while implementing big data analytics.
8. Write short notes on any one domain application of big data analytics.
9. How do business drivers influence the adoption of big data solutions?
10. Explain the importance of big data analytics in decision making.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

ANSWERS:

1. Define Big Data and explain why traditional data processing approaches are often insufficient.

Big Data refers to extremely large, complex, and rapidly generated datasets that cannot be efficiently stored, processed, or analyzed using traditional data-processing tools.

Traditional approaches are often insufficient because:

- They cannot handle **huge data volumes**.
- They have difficulty processing **high-speed data**.
- They mainly work with **structured data**, while big data also contains unstructured and semi-structured data.

- Scaling traditional systems can be expensive and time-consuming.
- Big data requires **distributed processing** across multiple computers.

Example: Social media platforms generate millions of posts, images, videos, and comments every day, which traditional databases cannot efficiently process alone.

2. Explain the 5Vs of Big Data with one practical example for any two Vs.

The **5Vs of Big Data** are:

1. **Volume** – Refers to the huge amount of data generated.
 - Example: Google stores enormous amounts of search and user data.
2. **Velocity** – Refers to the speed at which data is generated and processed.
 - Example: Stock market systems receive and process thousands of transactions per second.
3. **Variety** – Refers to different types of data such as text, images, audio, video, and numerical data.
4. **Veracity** – Refers to the accuracy, reliability, and quality of data.
 - Example: Fake or duplicate customer reviews can reduce data reliability.
5. **Value** – Refers to the useful information and benefits obtained from data analysis.

3. Differentiate structured, semi-structured, and unstructured data.

Type	Meaning	Examples
Structured	Data organized in a fixed format such as rows and columns	SQL tables, bank records, employee records
Semi-structured	Data does not follow a strict table structure but contains tags or labels	JSON, XML, emails
Unstructured	Data has no predefined structure	Images, videos, audio, social media posts

Example:

A customer database is structured data, a JSON customer record is semi-structured, and a customer review containing an image and video is unstructured data.

4. Explain the role of distributed file systems in big data processing.

A **Distributed File System (DFS)** stores large datasets across multiple computers or nodes in a network instead of storing everything on one machine.

Its major roles are:

- **Distributed storage:** Splits large files into smaller blocks and stores them across multiple nodes.
- **Fault tolerance:** Creates multiple copies of data so it is not lost if one node fails.
- **Scalability:** More machines can be added when the amount of data increases.
- **Parallel processing:** Multiple machines can process different parts of data simultaneously.
- **High availability:** Data remains accessible even when some machines fail.

Example: HDFS (Hadoop Distributed File System) is widely used for storing and processing big data.

5. List any three major security concerns in big data platforms.**1. Data Privacy**

Sensitive information such as customer details, financial information, and personal records must be protected from unauthorized access.

2. Unauthorized Access

Attackers or unauthorized users may access big data systems. Strong authentication and access control are required.

3. Data Security and Integrity

Data can be modified, deleted, or corrupted by attackers or system failures. Encryption, backups, and integrity checks help protect data.

6. What are data appliances and integration tools? Explain their role in big data workflows.

Data appliances are specialized hardware or software systems designed to store, manage, and process large amounts of data efficiently.

Integration tools are software tools used to collect data from different sources, transform it, and move it into a common storage or processing system.

Role in big data workflows:

Data Sources → Integration Tools → Data Storage/Appliance → Processing → Analytics → Decision Making

They help to:

- Collect data from multiple sources.
- Clean and transform data.
- Move data between different systems.
- Improve processing performance.
- Make data available for analytics.

7. Discuss two major challenges organizations face while implementing big data analytics.

1. Data Quality and Management

Big data comes from many sources and may contain duplicate, incomplete, incorrect, or inconsistent information. Organizations need proper data cleaning and management techniques.

2. Security and Privacy

Big data often contains sensitive customer and business information. Organizations must protect it from unauthorized access, cyberattacks, and misuse while complying with privacy regulations.

Other challenges include high implementation costs, lack of skilled professionals, and infrastructure complexity.

8. Write short notes on any one domain application of big data analytics.

Big Data in Healthcare

Big data analytics is widely used in the healthcare sector to analyze patient records, medical images, laboratory results, and other healthcare data.

Applications include:

- Predicting diseases and health risks.
- Improving patient treatment.
- Monitoring patients in real time.
- Analyzing medical research data.
- Reducing hospital costs.
- Improving hospital resource management.

Example: Hospitals can analyze historical patient data to identify patients who may be at high risk of developing a particular disease.

9. How do business drivers influence the adoption of big data solutions?

Business drivers are the business needs and goals that encourage organizations to adopt big data technologies.

Major drivers include:

- **Better decision making:** Data analytics provides useful insights for managers.
- **Customer understanding:** Companies can analyze customer behavior and preferences.
- **Cost reduction:** Analytics can identify unnecessary expenses and improve resource usage.
- **Competitive advantage:** Organizations can use data to develop better products and services.
- **Revenue growth:** Data can help identify new markets and business opportunities.
- **Operational efficiency:** Businesses can optimize processes using real-time information.

Thus, business requirements and the need to gain value from data strongly influence the adoption of big data solutions.

10. Explain the importance of big data analytics in decision making.

Big Data Analytics helps organizations examine large amounts of data and extract useful patterns, trends, and insights.

Its importance in decision making includes:

- Provides **accurate and data-driven decisions**.
- Helps identify **market trends**.
- Improves **customer satisfaction**.
- Helps predict future events and risks.
- Reduces uncertainty in business decisions.
- Improves operational efficiency.
- Helps organizations gain a **competitive advantage**.

Example: An online shopping company can analyze customer purchase history to recommend products and predict future demand.

Conclusion: Big data analytics converts raw data into useful information, helping organizations make faster, smarter, and more informed decisions.